Data-Rich, Compute-Poor: Why India Is Behind in the AI Race

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By Pratik Priyaranjan Sahoo, CFA Level II Candidate · Reading time: ~8 minutes

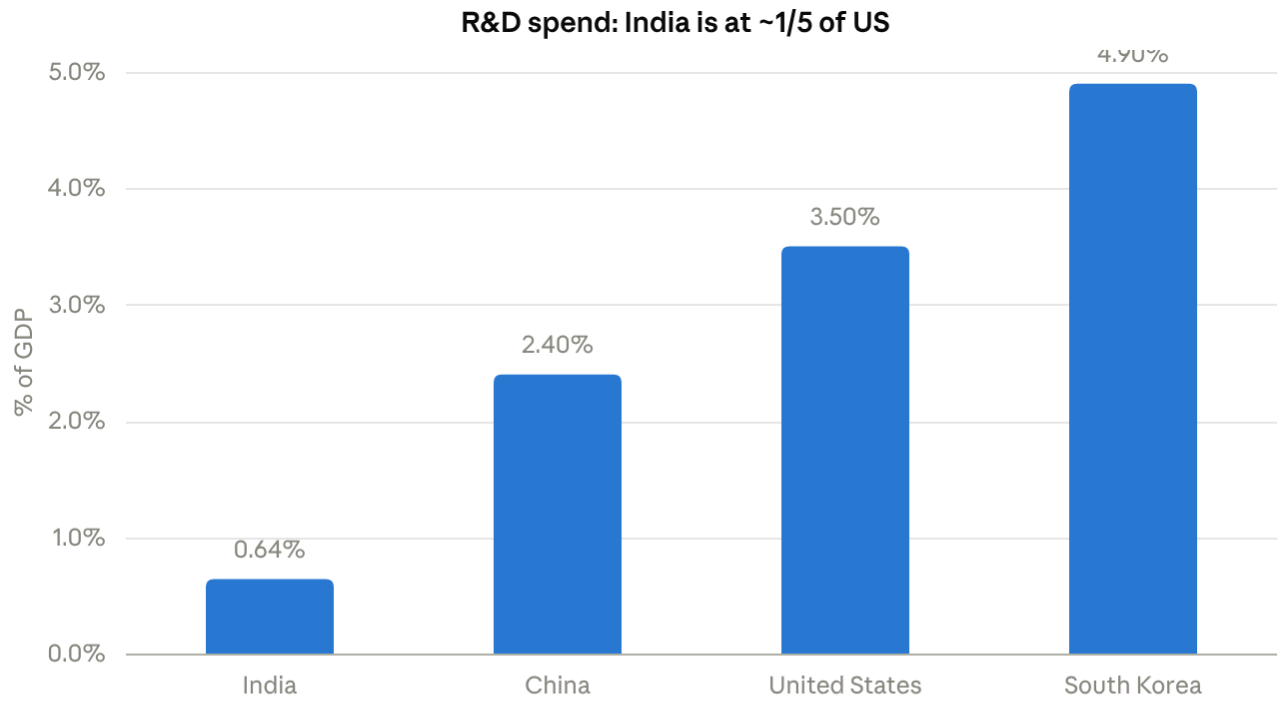
India generates close to a fifth of the world's data, has one of the largest pools of engineers on the planet and hosted a global AI summit this year that drew over $200 billion in pledges. Yet when you look at who is building frontier models, who owns the compute and who captures the profits, India is still a spectator more than a player. This piece looks at why — and whether the gap can close.

The scoreboard

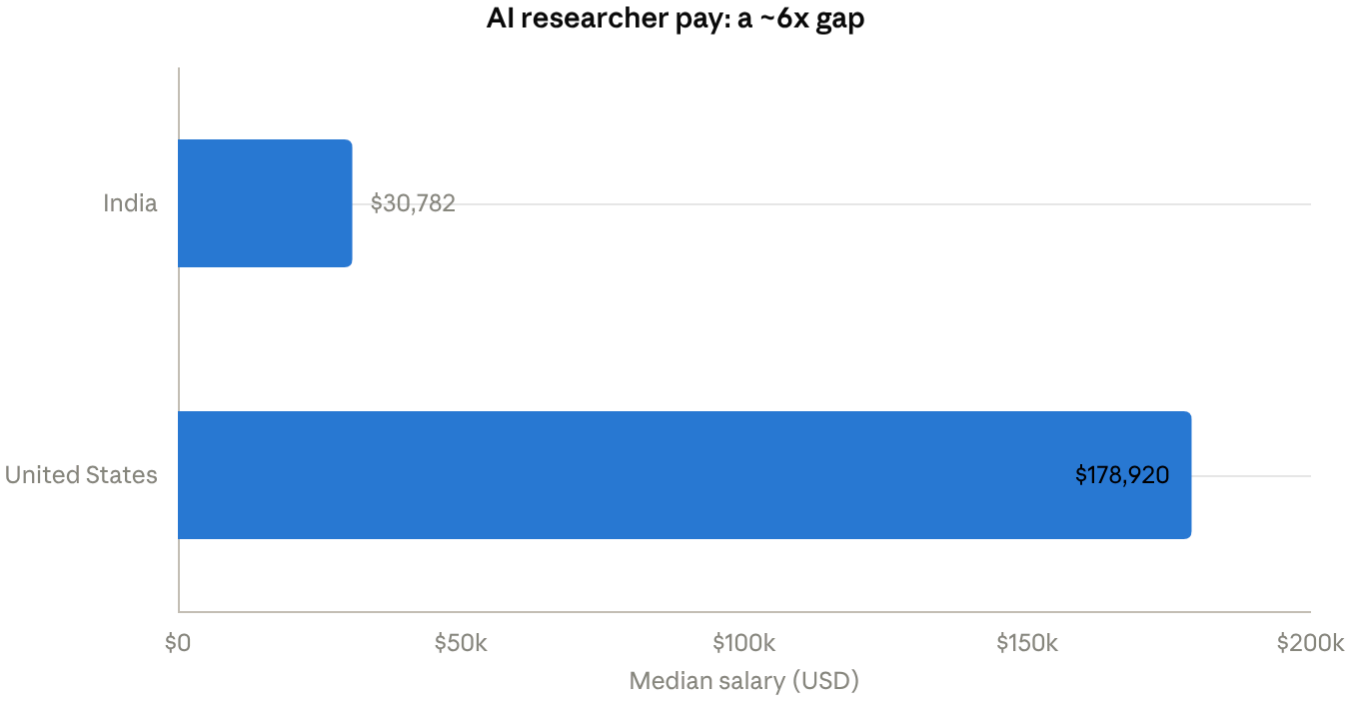
The gap is not about ambition; it is about scale. On every hard metric that decides who wins in AI — research spending, compute, capital and pay — India is operating at a fraction of the leaders.

Metric	India	United States	China
R&D spend (% of GDP)	~0.64% (Economic Survey 2025–26)	~3.5%	~2.4–2.5%
Private sector share of R&D	~36%	~75%	~77%
AI infrastructure spend (% of GDP, 2026)	Negligible	~2%	~0.4%
Flagship public AI programme	IndiaAI Mission: ₹10,372 crore (~$1.25 bn)	Private hyperscalers guiding ~$725 bn of 2026 capex	State-backed plus Big Tech
Median AI researcher pay	~$30,800	~$178,900	—

To put the IndiaAI Mission in perspective: its entire multi-year budget is smaller than what a single US hyperscaler spends on data centres in a couple of weeks. The US alone accounts for more than 80% of an estimated $800 billion in global AI infrastructure spending this year.



Gross R&D expenditure as % of GDP, latest available estimates.



Median annual salary of an AI researcher, US dollars.

Economic Survey 2025–26, CareEdge Ratings, Lapaas Voice · 2026

Six reasons India is behind

1. India under-invests in research — and the private sector barely participates

India's gross R&D spending has been stuck at roughly 0.6–0.7% of GDP for years, and even the most recent upward revision keeps it below 1%. The Economic Survey 2025–26 is candid that the problem is not talent. It is that Indian companies contribute only about a

third of national R&D, and research rarely crosses the "valley of death" from lab prototype to commercial product. CareEdge Ratings adds that corporate R&D is concentrated in a few sectors like autos and pharma, and tends to be incremental rather than breakthrough.

Frontier AI is the most R&D-intensive industry in history. A country that spends 0.6% of GDP on research cannot expect to build models that compete with labs spending tens of billions a year.

2. Compute: the new oil, and India is importing it

India is data-rich but compute-poor. The IndiaAI Mission's public GPU pool has grown from about 34,000 to a targeted 54,000, with a goal of 100,000 GPUs by December 2026. That sounds large until you remember that a single frontier training run at a US lab can use more chips than India's entire public pool.

There is also an execution gap. Reporting in April 2026 suggested only around ₹400 crore of the ₹10,372 crore mission had actually been released, and less than half of the FY26 allocation was spent. Announcements are running well ahead of deployment.

3. Capital is patient in the US and impatient in India

Global venture capital hit about $510 billion in the first half of 2026 alone, with OpenAI and Anthropic absorbing 43% of it. Indian venture money, by contrast, remains weighted toward SaaS, fintech and consumer apps — businesses with predictable, near-term returns. Foundation-model companies burn enormous sums for years before revenue arrives, and India has few investors willing to underwrite that. In a Blind survey of 1,723 Indian tech professionals, 12% named the lack of long-term risk capital as the main barrier.

4. The talent is Indian, but the jobs are abroad

India ranks at or near the top globally in AI skill penetration, and Indian-origin researchers are everywhere in Silicon Valley. The problem is where they work. A median AI researcher in the US earns about $178,900 a year versus roughly $30,800 in India — nearly a six-fold gap, before counting access to world-class compute and frontier problems. The new Prime Minister Research Chair scheme to attract researchers back is a start, but money and GPUs, not grants alone, move people.

5. Every Indian model runs on American chips

Sarvam's models, BharatGen's Param2 and the national GPU pool all depend on NVIDIA hardware. India's semiconductor push is real, but it is years from producing AI-grade chips. That leaves India's "sovereign AI" exposed to US export policy, and there is still no binding long-term guarantee of access to top-end chips that would survive a change of administration in Washington.

6. A services mindset in a products race

India's tech success story was built on IT services: selling skilled hours to global clients. That model created wealth and jobs, but it trained the ecosystem to implement others' technology rather than own the intellectual property. Tellingly, 62% of respondents in the Blind survey cited the absence of homegrown R&D and IP as India's primary barrier, and only 5% saw India as a global AI leader. Worse, generative AI now threatens the services model itself by automating the very tasks India's IT and BPO firms sell.

The counter-argument: India is not out of the race

It would be lazy to call India a lost cause. There are real strengths:

- **Indigenous models are shipping.** At the India AI Impact Summit in February 2026, Sarvam AI launched 30-billion and 105-billion parameter models trained on IndiaAI Mission compute and built for Indian languages. The government-backed BharatGen Param2, a 17-billion parameter model covering 22 languages, launched alongside.
- **Investment pledges are huge.** The summit drew more than $200 billion in commitments across data centres, models and applications, from Reliance, Adani and global hyperscalers.
- **Digital public infrastructure is a unique asset.** Aadhaar, UPI and ONDC give India a population-scale platform to deploy AI in payments, identity and public services that few countries can match.
- **Adoption, not invention, may be the prize.** IT Minister Ashwini Vaishnaw has pushed back on the "lagging" narrative, calling this the first chapter of the AI revolution. There is a credible argument that India's biggest economic gains will come from applying AI across 1.4 billion people, not from winning the frontier-model race.

The honest conclusion: India is behind in *building* AI, but it may not be behind in *using* it.

Why this matters for investors

The AI gap is not just a technology story; it is showing up in the stock market. A Reuters poll of 24 analysts in May flagged India's limited participation in the global AI-led rally as one of the reasons the Nifty and Sensex could post their first annual loss since 2015. When global money is chasing AI winners in the US, Taiwan and Korea, a market dominated by banks, consumer names and IT services looks less exciting — and India's IT exporters face the added risk that AI compresses their billable-hours model.

My view: what would actually close the gap

1. **Deploy what has been announced.** Speeding up disbursement of the IndiaAI Mission budget matters more than announcing new headline numbers.
2. **Pay for talent, not just grants.** Competitive salaries plus guaranteed compute access are what bring researchers home.
3. **Force private R&D up.** Stronger tax incentives and procurement preferences for Indian IP could push corporate R&D toward the ~2% of GDP that CareEdge suggests by 2035.
4. **Build patient capital.** A sovereign deep-tech fund or venture-debt programme, similar to Israel's and Singapore's models, would fund the long-gestation bets private VCs avoid.
5. **Solve power and chips together.** Data centres need reliable, cheap electricity, and India needs long-term chip supply agreements while its fabs mature.
6. **Play to strengths.** Indic-language models, DPI-integrated AI and applied AI for agriculture, health and finance are races India can realistically win.

Bottom line: India is not behind because it lacks brains or data. It is behind because the AI race is a capital-intensive, research-heavy, hardware-dependent contest — and on all three counts India started late and is spending small. The next two to three years of execution will decide whether India becomes an AI producer or remains the world's largest AI consumer.

Disclaimer: This article reflects the author's personal analysis for educational purposes and is not investment advice. Data as of 27 September 2026.

Sources

- Forbes India — India's R&D expenditure just 0.6% of GDP: Economic Survey
- Business Today — CareEdge: India's R&D spend should rise to 2% of GDP
- Kapyn — The IndiaAI Mission, explained
- IFB Trend — IndiaAI Mission: 100,000 GPUs by 2026
- explainx.ai — India sovereign AI status 2026
- American Bazaar — Most Indian tech professionals say India is lagging in AI
- Lapaas Voice — India wants AI talent back, but what's the incentive?
- Memeburn — AI global funding statistics 2026
- Wikipedia — India AI Impact Summit 2026
- Whalesbook — India's AI dilemma: huge talent, missing capital
- Business Today — Sensex, Nifty may deliver negative returns in 2026 (Reuters poll)